

Queue Class - 2

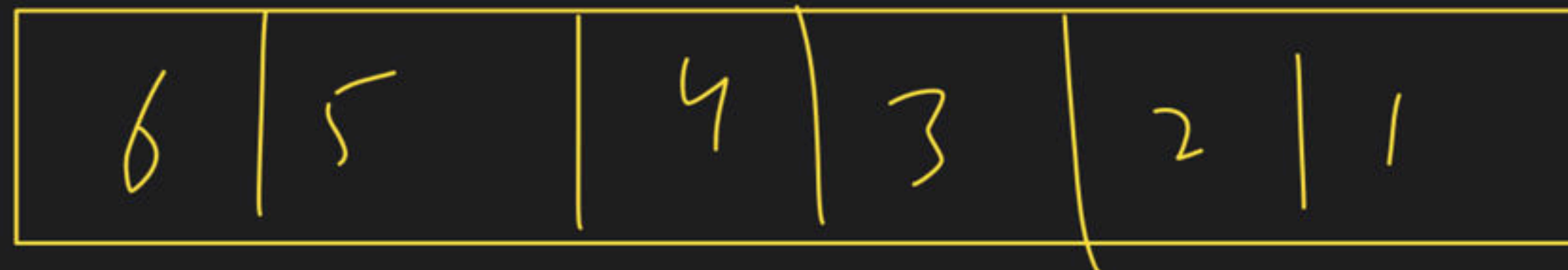
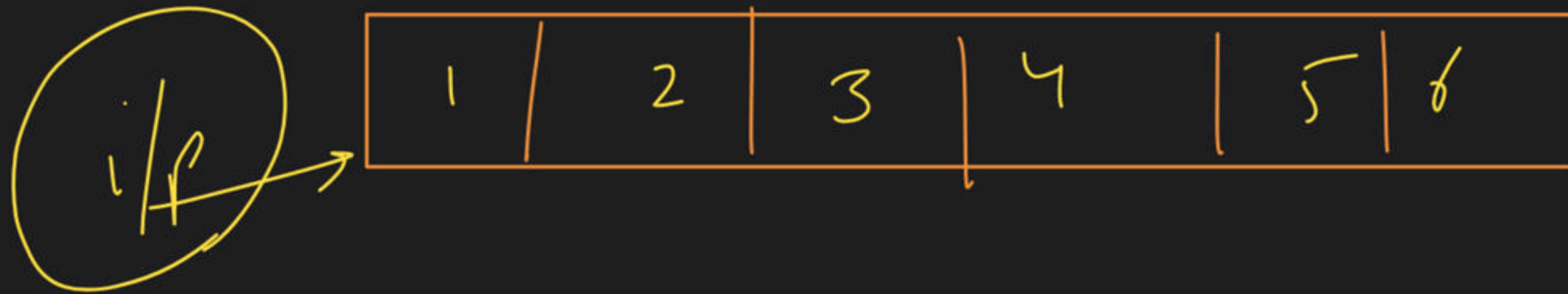
Special class

→ Reverse a Queue

STL

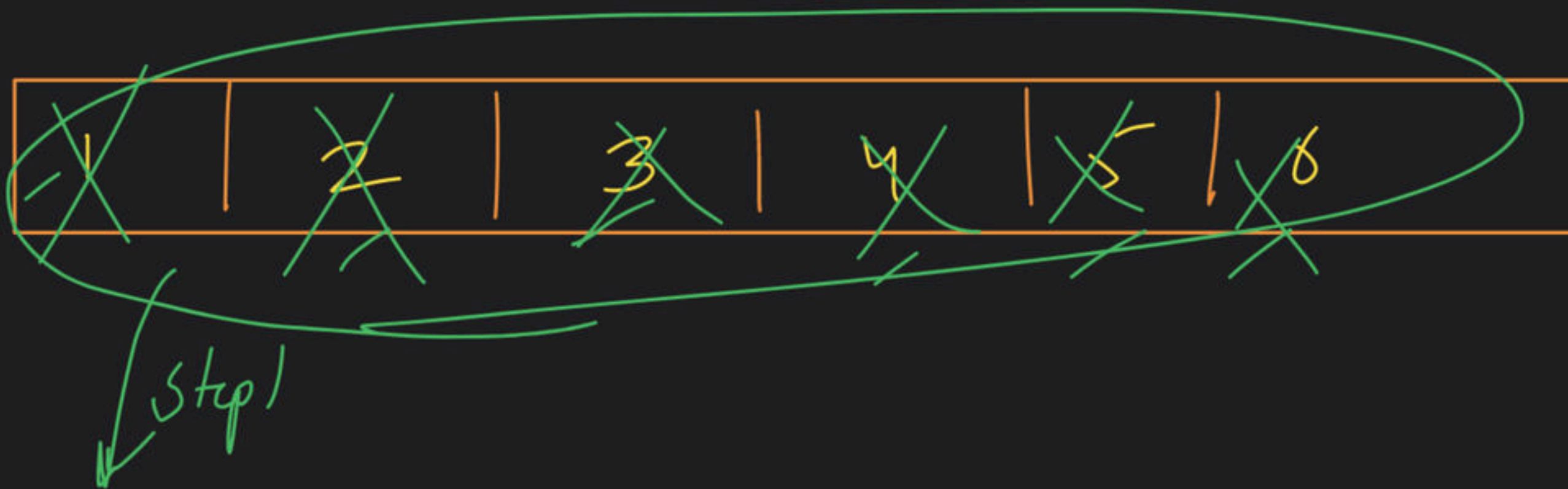
→ Rec //

→ Stack //



o/p

2

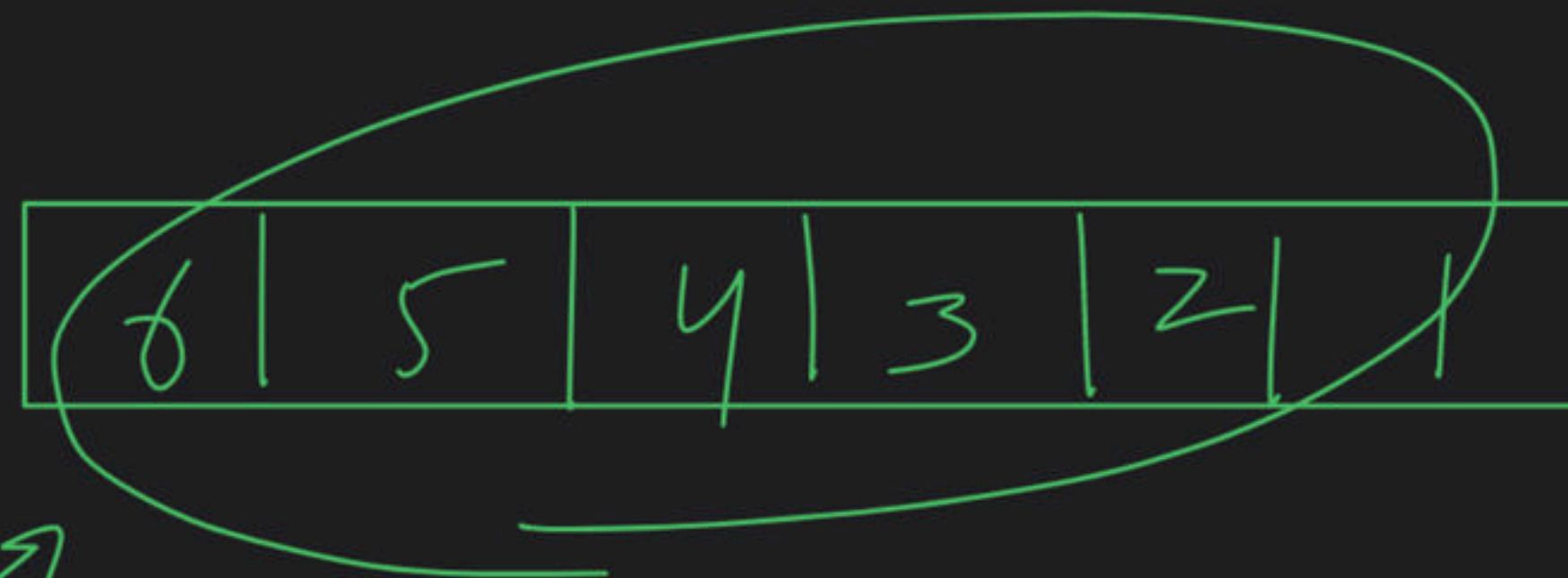


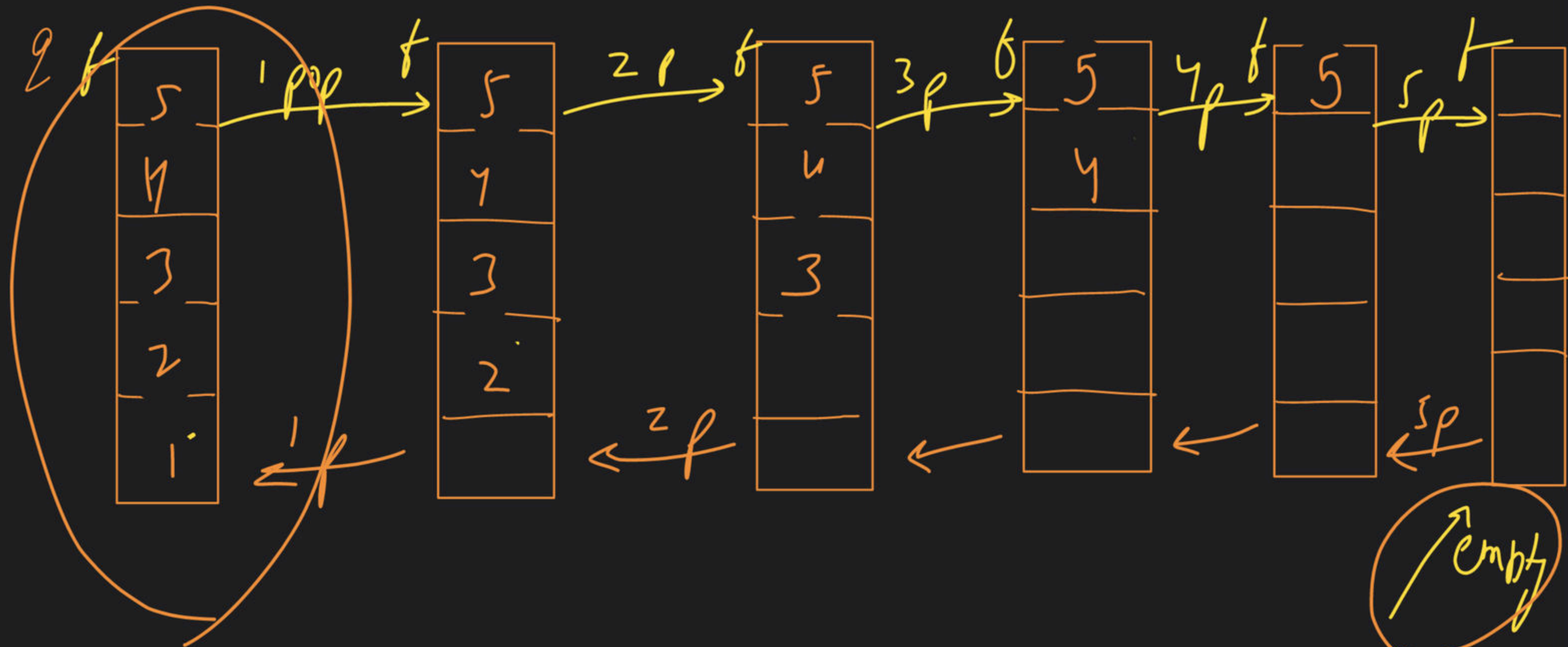
Stack



Step 2

2



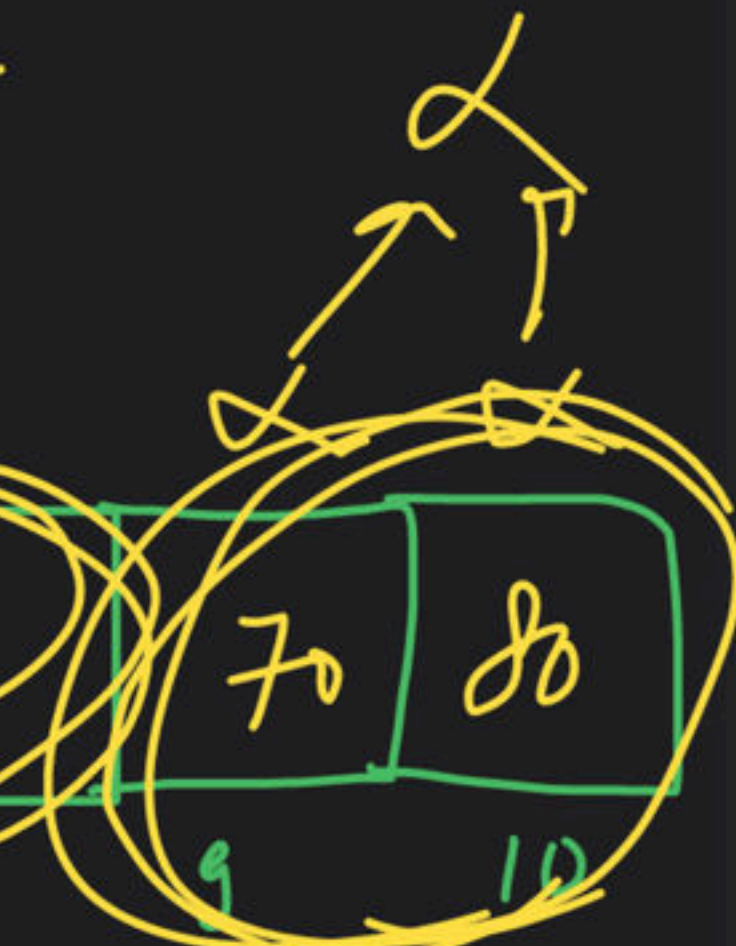
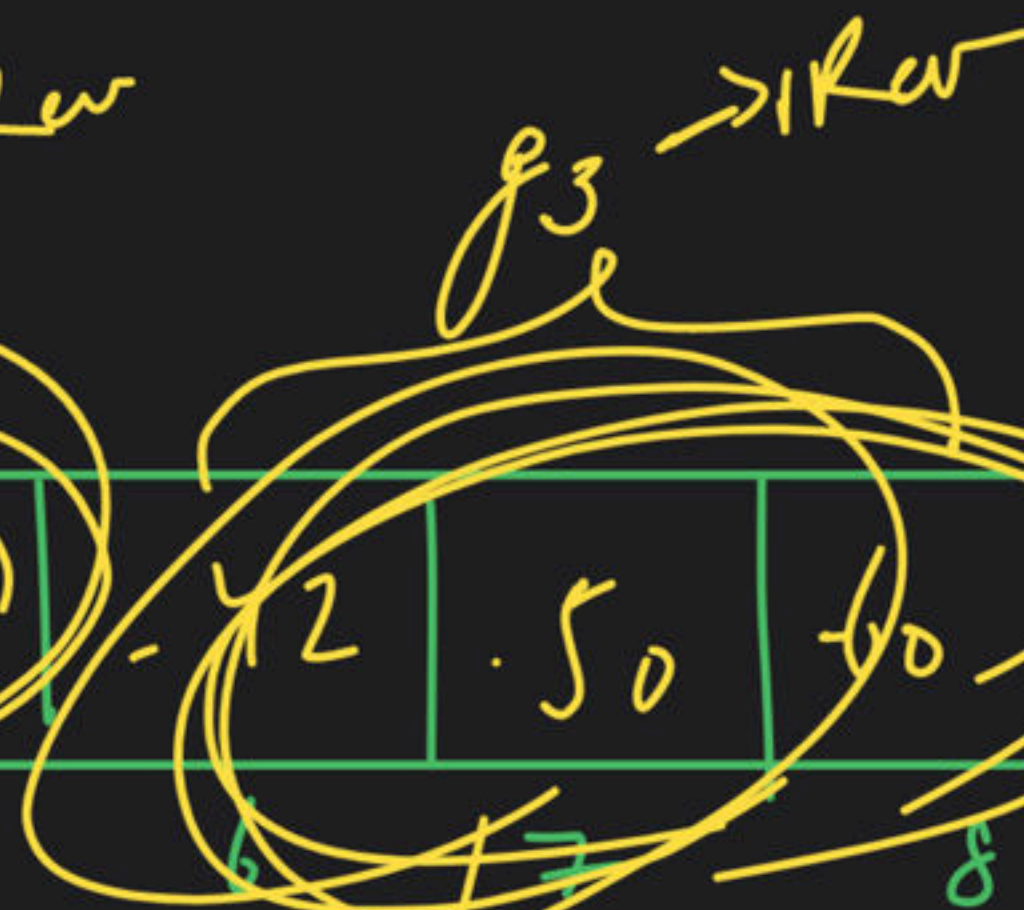
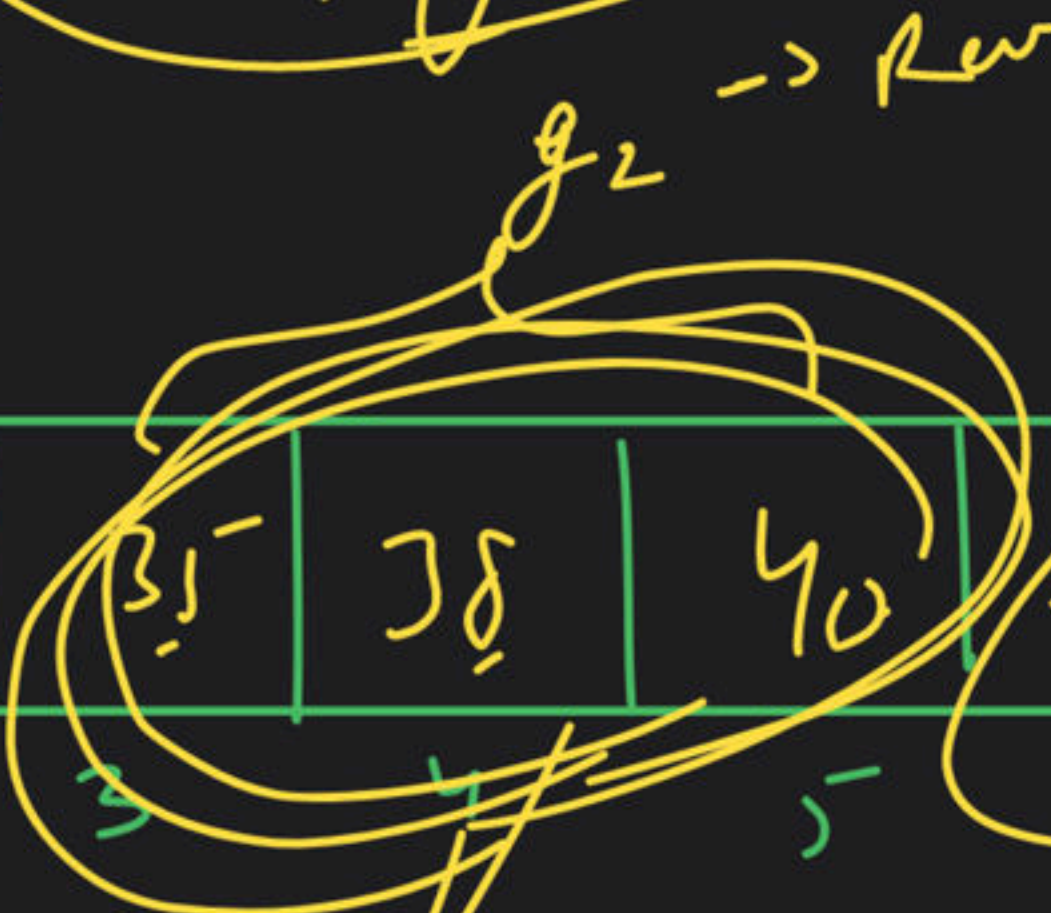
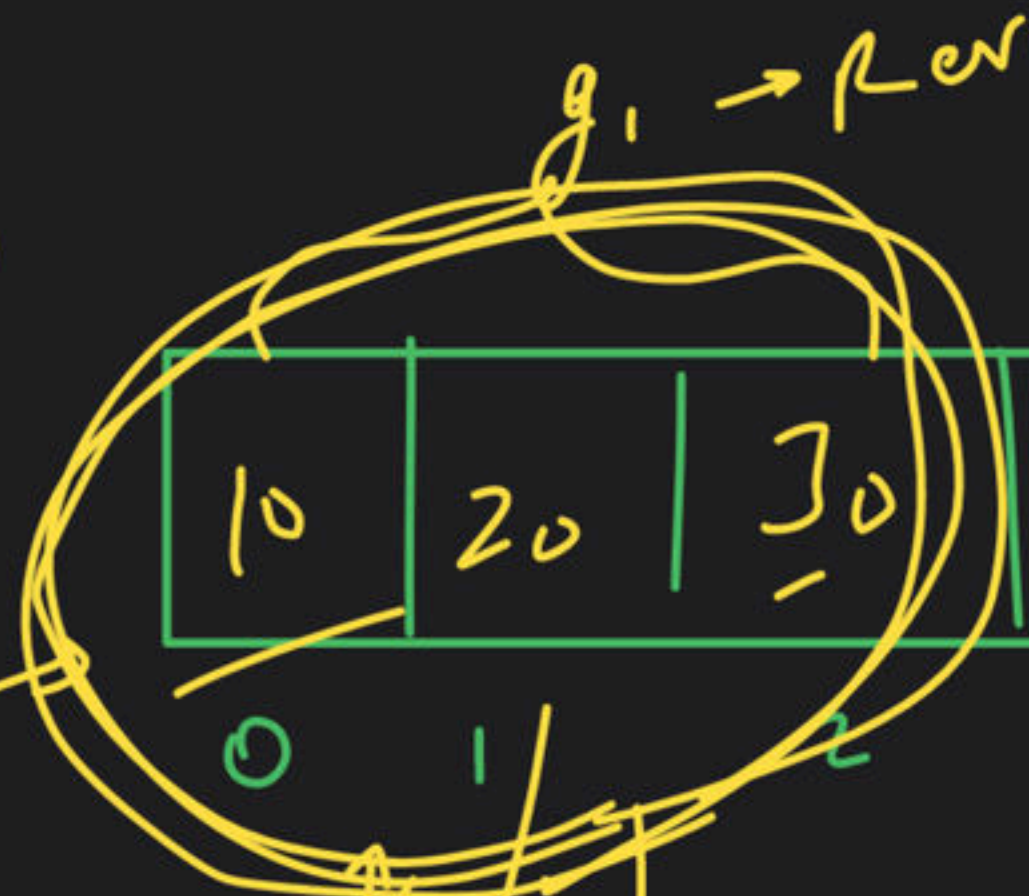


empty

B.C

→ Reverse in K group

$K=3$



o/p

30	20	10	80	50	40	90	80	70	100	110
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→ Queue

→ Pattern

3 step

2 minute

size = 11

10	20	30	40	50	60	70	80	90	100	110	30	20	10
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2

(K)

$11 - 3 \rightarrow 8$

40	50	60	70	80	90	100	110	30	20	10	60	50	40
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$8 - 3 \rightarrow 5$

70	80	90	100	110				90	80	70
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30	20	10
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60	50	40
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90	80	70
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$5 - 3 \Rightarrow 2 < 12$

100	110	30	20	10	60	50	70	90	80	70	100	110
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$k=3$ size $\Rightarrow 11$

Diagram 1: Initial array of size 11

10	20	30	40	50	60	70	80	90	100	110
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Annotations: q points to the first element (~~10~~). st points to the third element (~~30~~). A bracket groups the first three elements (~~10~~, ~~20~~, ~~30~~).

size $= 11 - 3 = 8$

Diagram 2: Array after removing first 3 elements

40	50	60	70	80	90	100	110	30	20	10
---------------	---------------	---------------	----	----	----	-----	-----	----	----	----

Annotations: q points to the first element (~~40~~). st points to the third element (~~60~~). A bracket groups the first three elements (~~40~~, ~~50~~, ~~60~~).

size $= 8 - 3 = 5$

Diagram 3: Array after removing first 3 elements

70	80	90	100	110	30	20	10	60	50	40
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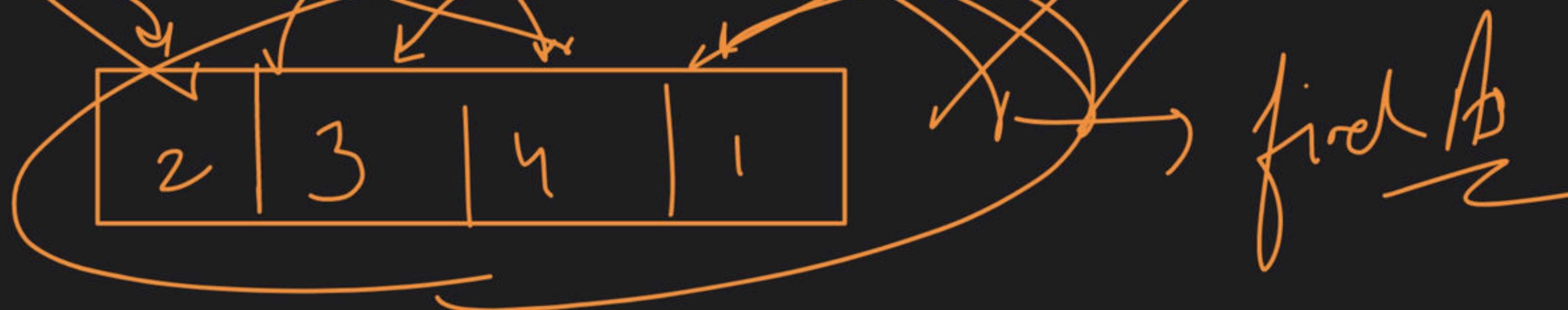
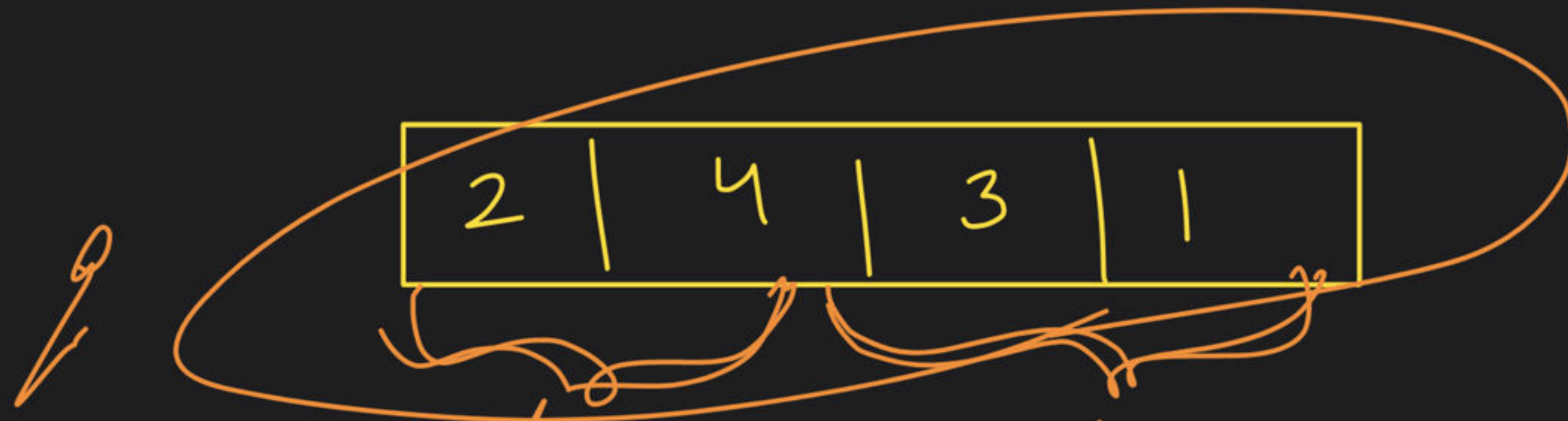
Annotations: q points to the first element (~~70~~). st points to the third element (~~90~~). A bracket groups the first three elements (~~70~~, ~~80~~, ~~90~~).

size $= 5 - 3 = 2 < k$

Diagram 4: Array after removing first 3 elements

100	110	30	20	10	60	50	40	90	80	70
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Annotations: q points to the first element (~~100~~). st points to the second element (~~110~~). A bracket groups the first two elements (~~100~~, ~~110~~). A curved arrow points from the top of the array to the bottom element (70). A bracket at the bottom groups the last three elements (90, 80, 70).



second

g₁



g₂



first

Q1

Q1



↓ step 1 → first half → queue

↗↘ Second Half

Q2



← First Half

ans



Reverse

→ Sliding Window Pattern



bundle along
s²

K=2

First window

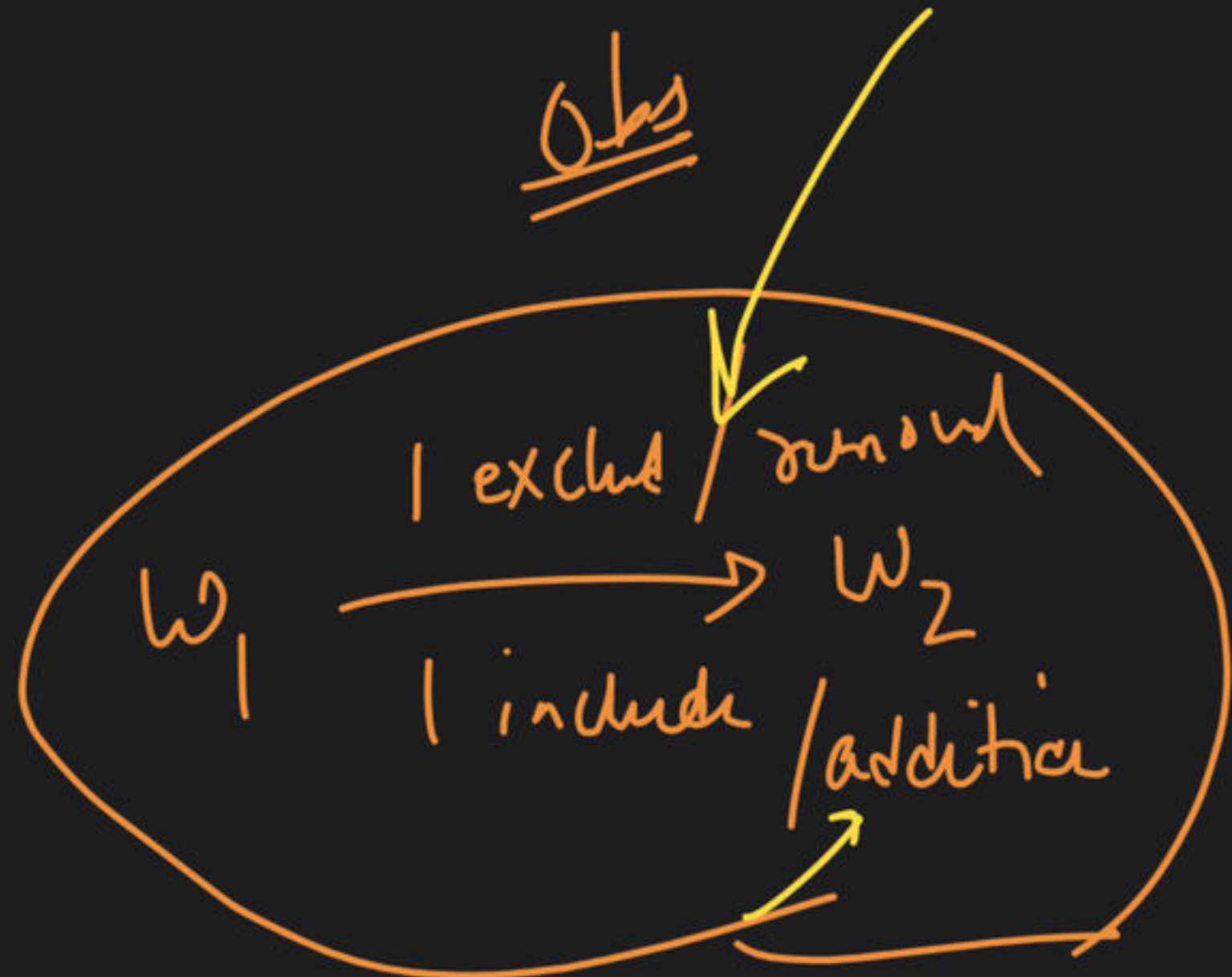
-8	2	3	-6	10
0		2	3	4

-8

0

-6

-6



3 step → SW

→ ① Process First Window

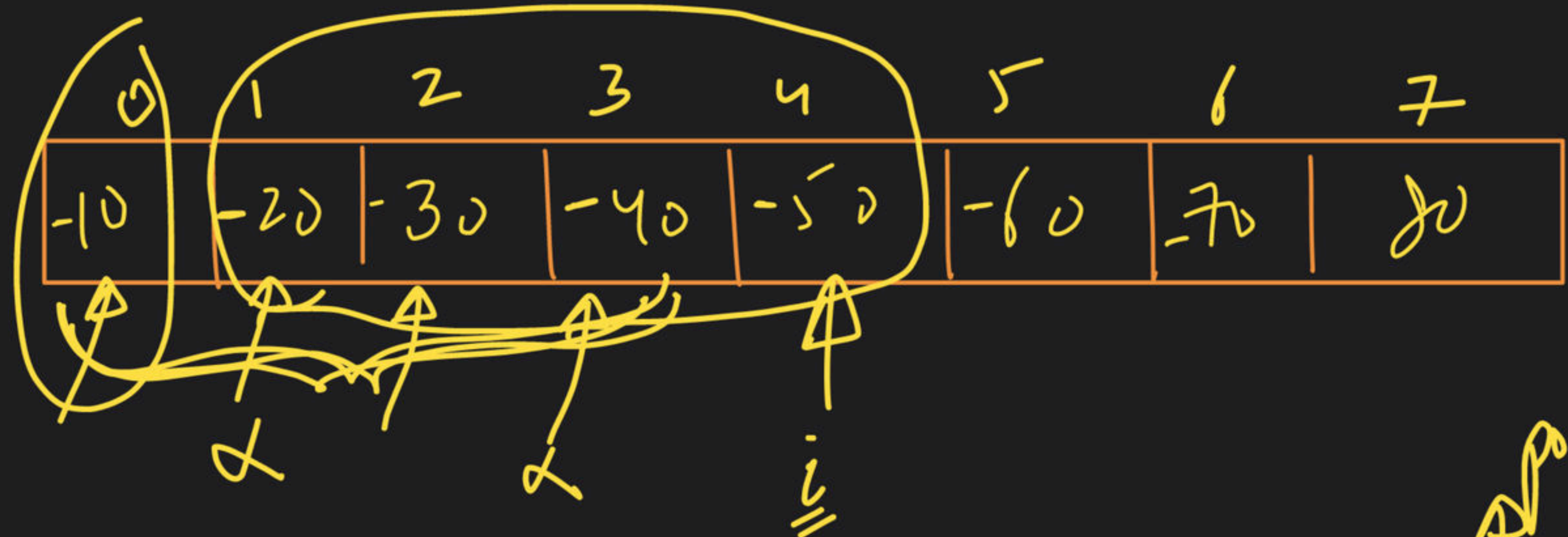
→ ② Process remaining window

→ ans

→ removal

→ addition

K=4



$$4 - 0 = 4$$

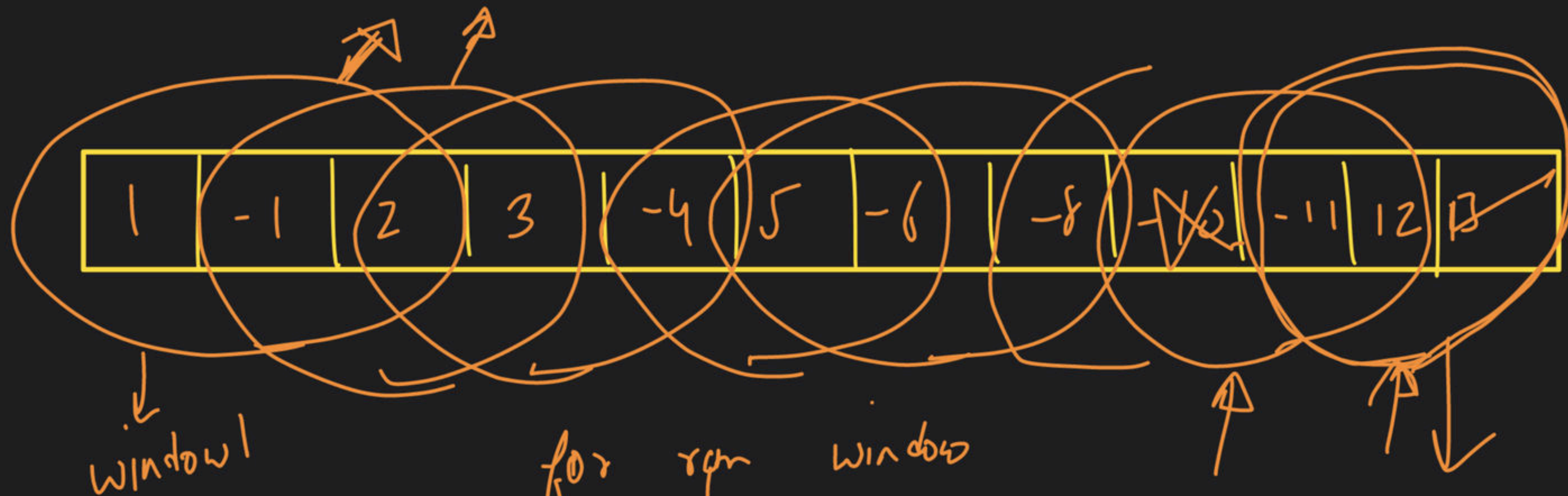
$$4 - 1 = 3$$

$$4 - 2 = 2$$

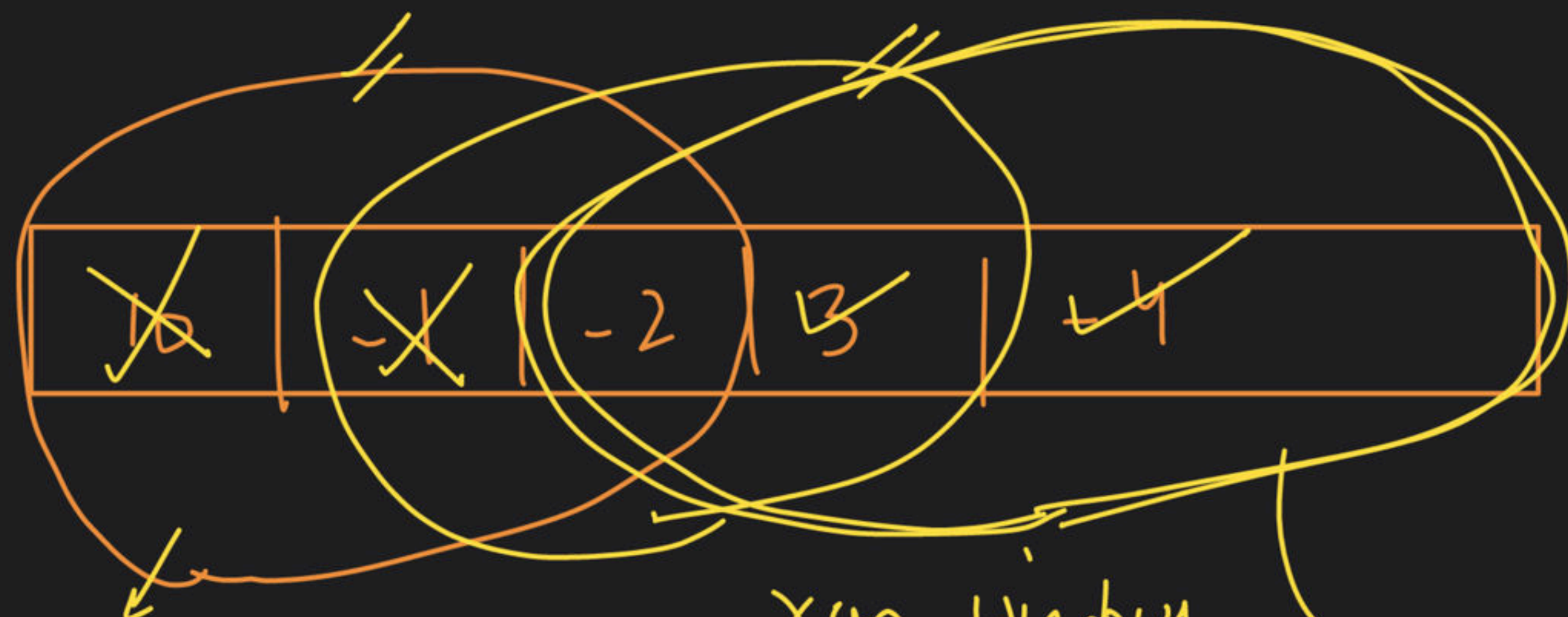
$$4 - 3 = 1$$

Out of range
P

$$i - dq.front() \geq K$$

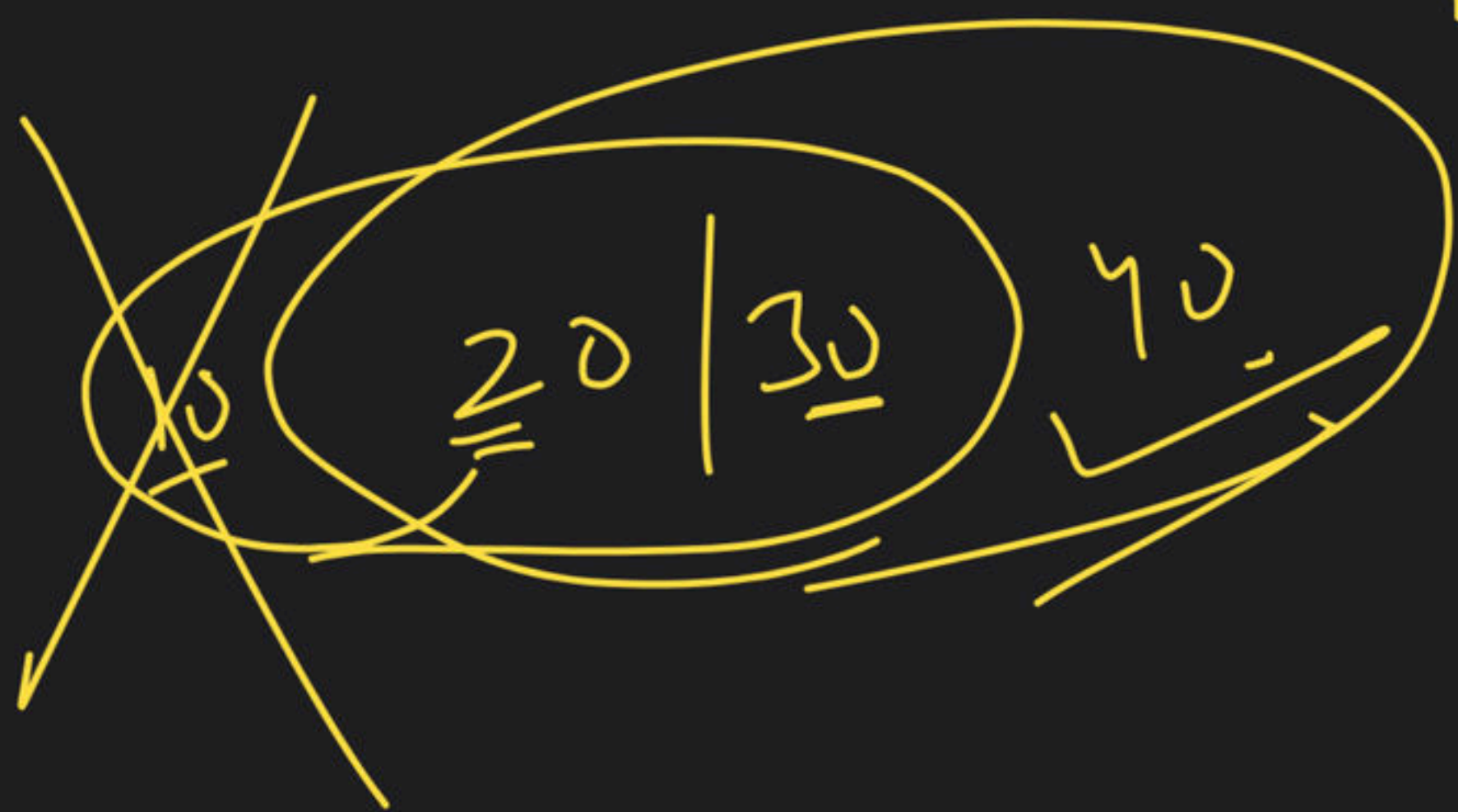


$k=3$

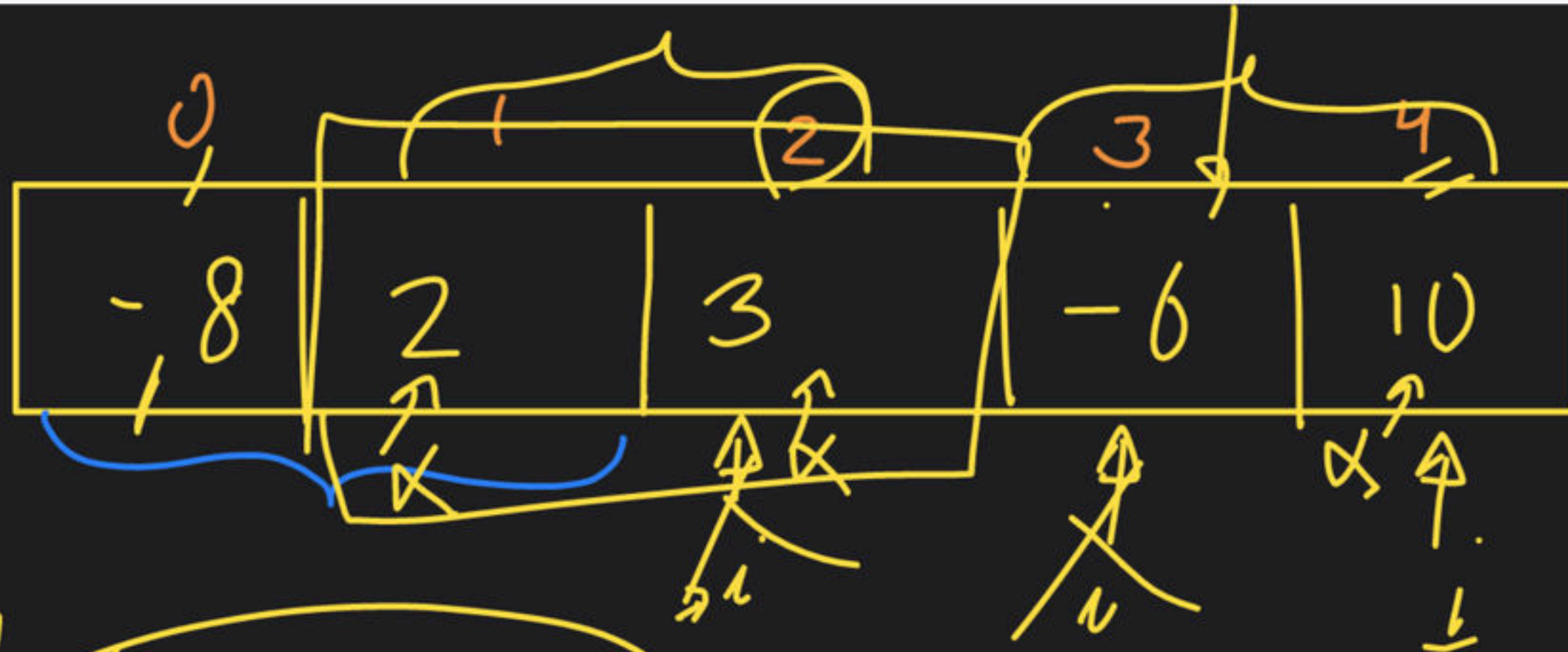


w_1

your window
└─┘

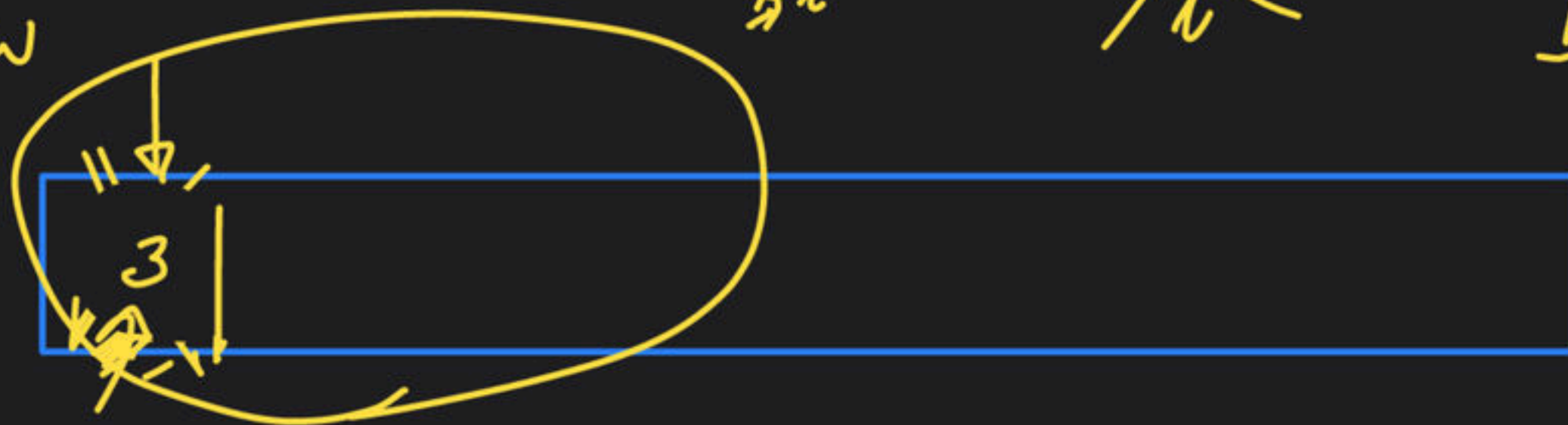


w_2



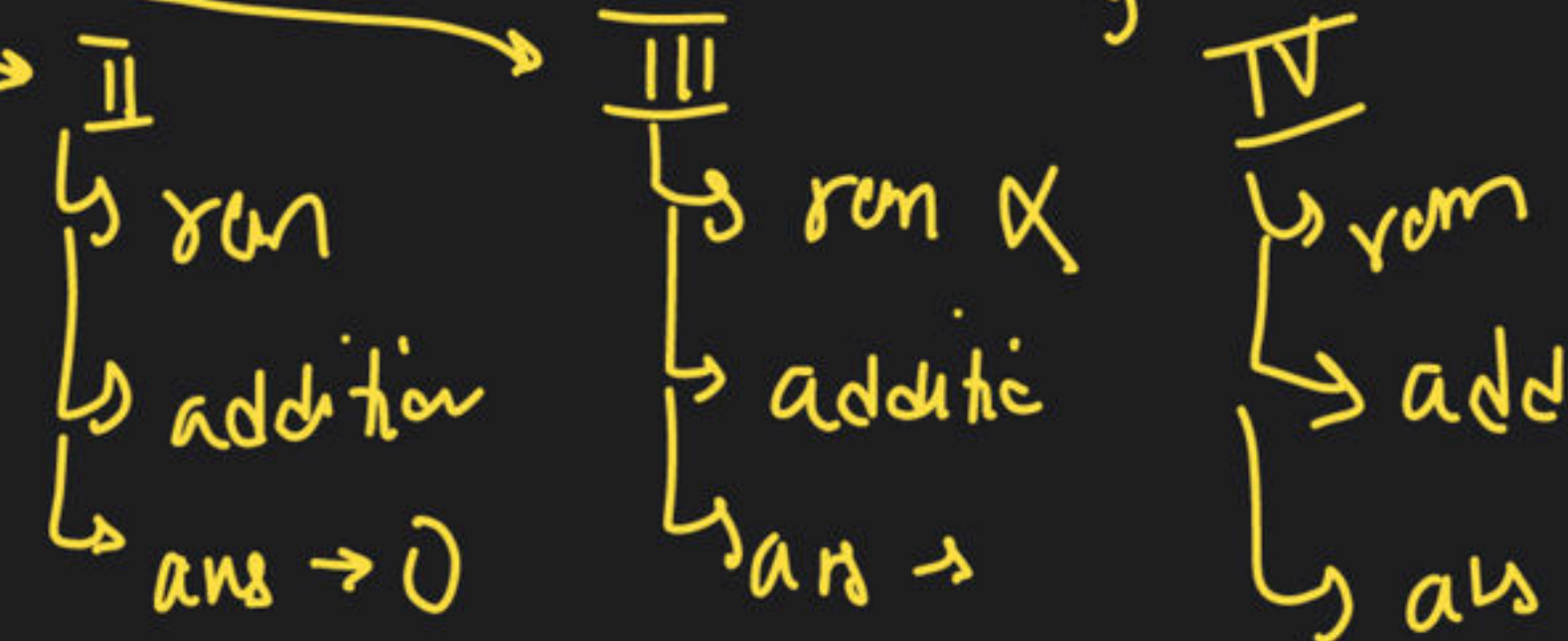
$K=2$

① Ist window



Ist window $\rightarrow$ ans

② Rem Window



81

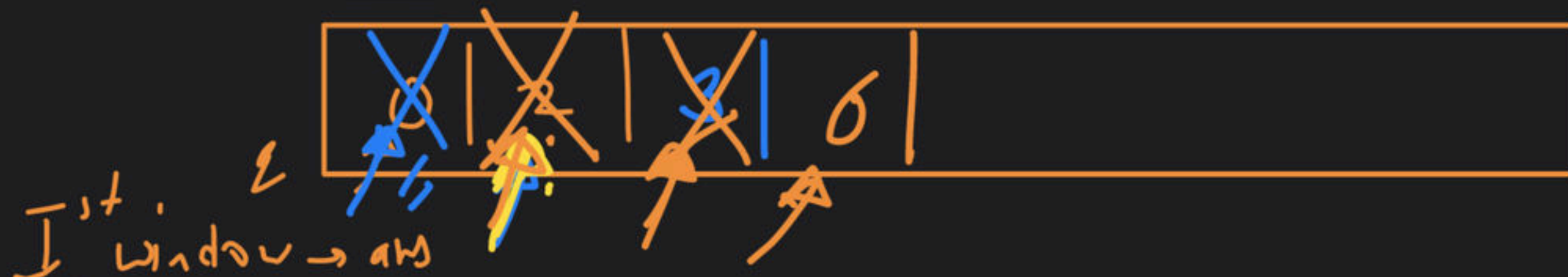
ans

-8
0
-6
-6

K=3



1st window passes



1st window -> ans

sum window

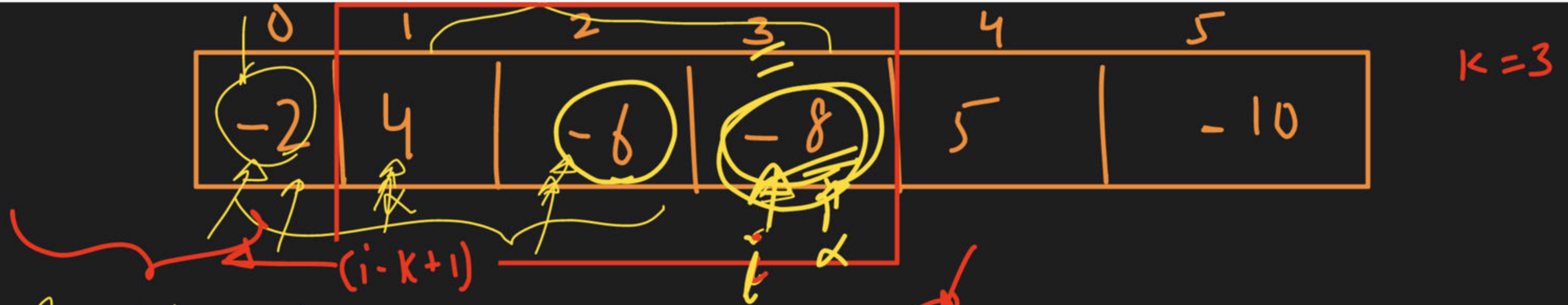
II
└ sum
└ add
└ ans

III
└ sum
└ add
└ ans

IV
└ sum
└ add
└ ans

V
└ sum
└ add
└ ans

-1
-4
-5
-10



(2) Remaining window $\rightarrow$ proxy



(II) $\rightarrow$ removal $\rightarrow$ current window $\rightarrow (i-k+1) \rightarrow i$

$$dq.front() < i-k+1$$

$\rightarrow$ remove krdo

$\rightarrow$ additn $\rightarrow$ -ve $\rightarrow$ index jth

$\rightarrow$ ans

Kal →

Extra Class

8:30

Ques

Complete



















